DETAILED DIAGNOSTIC REPORT

COMPREHENSIVE PROPERTY ANALYSIS & ASSESSMENT

■ AI-Powered Professional Analysis

■ Generated on:	March 18, 2026 at 10:59 PM
■ Report Type:	Premium DDR Analysis
■ Analysis Method:	Advanced AI Technology
■■ Images Processed:	Inspection: N/A, Thermal: N/A
■ Processing Engine:	Google Gemini 2.5 Flash

Professional Disclaimer: This premium report has been generated using advanced AI analysis of inspection and thermal reports. While our AI system provides comprehensive insights, all findings should be verified by qualified professionals before taking any action. This report represents a sophisticated analysis tool to support expert decision-making.

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**Page numbers are approximate and may vary based on content length*

1. PROPERTY ISSUE SUMMARY

This comprehensive diagnostic assessment reveals a residential building in Kolkata experiencing notable issues primarily concerning moisture intrusion and compromises in its thermal envelope. The most critical immediate concern involves active water leakage coupled with visible mold growth within the bedroom, findings that are strongly corroborated by advanced thermal imaging analysis indicating pervasive moisture presence. Concurrently, the living room exhibits physical signs of distress, including visible wall cracks and paint peeling, which, when combined with thermal data suggesting significant temperature variations and heat leakage, points towards underlying structural considerations and inefficiencies in the building's thermal performance. These findings collectively necessitate urgent intervention to safeguard both the structural integrity of the property and the health of its occupants.

2. AREA-WISE OBSERVATIONS

Living Room:

A thorough visual inspection of the living room identified discernible cracking on multiple wall surfaces, suggesting potential structural movement or material fatigue. Furthermore, widespread paint peeling was observed, indicating a compromise in adhesion possibly due to surface preparation issues, age, or latent moisture effects. Complementing these visual findings, thermal imaging analysis revealed a high degree of temperature variation across the room, which, alongside indicators of possible heat leakage, points towards a significant breach in the thermal resistance or air barrier integrity of the building envelope in this area.

Bedroom:

Direct and undeniable evidence of water leakage was noted on the ceiling within the bedroom, signifying active water penetration. This moisture intrusion has led directly to the detection of visible mold growth, highlighting a sustained period of damp conditions. Thermal imaging provided crucial corroborative evidence, precisely identifying cool spots and a broader thermal anomaly within the ceiling area. These thermal signatures are unequivocally consistent with the presence of active moisture and ongoing water seepage, confirming the severity of the visual observations.

3. PROBABLE ROOT CAUSE

The pervasive water leakage and subsequent mold growth identified in the bedroom are most likely attributable to a significant external water intrusion event, originating from a compromised element such as the roofing system, an external wall defect, or a plumbing anomaly concealed within the ceiling assembly. The corroborating thermal data, revealing distinct cool spots, strongly reinforces the conclusion of active and ongoing moisture presence from an external source. In the

living room, the visible wall cracks and paint peeling, when considered in conjunction with the high temperature variations and probable heat leakage detected via thermal analysis, collectively suggest a combination of factors. These could include structural stresses leading to material fatigue or movement, exacerbated by deficiencies in the building's thermal insulation or air sealing, which permits uncontrolled heat transfer. The presence of moisture, even if not explicitly stated as the primary cause in the living room, can often exacerbate paint adhesion issues and contribute to material degradation.

4. SEVERITY ASSESSMENT

Bedroom: Critical

The active water leakage on the ceiling coupled with the established presence of mold growth constitutes a critical and immediate concern. Active water intrusion carries inherent risks of rapid structural deterioration, potential electrical hazards if wiring is compromised, and extensive property damage. Mold growth, a direct biological consequence of prolonged moisture, presents significant health risks, particularly for individuals with respiratory sensitivities, and its proliferation demands immediate and specialized remediation to prevent further spread and health impacts. The definitive thermal evidence of active moisture substantiates this critical rating, indicating an ongoing and severe issue.

Living Room: High

The visible wall cracks and observable paint peeling, while sometimes appearing cosmetic, are indicators of underlying conditions that range from superficial surface distress to more substantial structural movement. When these visual cues are combined with thermal imaging revealing high

temperature variation and possible heat leakage, they collectively point to a significant compromise in the building envelope's integrity. This compromise results in energy inefficiency, potential discomfort for occupants, and, if the cracks are symptomatic of structural issues, could signify foundational movement or building settlement requiring urgent engineering assessment. The potential for both structural implications and ongoing energy loss elevates this condition to a high severity rating.

5. RECOMMENDED ACTIONS

Immediate Action (Bedroom):

It is critically imperative to promptly engage the services of a qualified roofing contractor, and potentially a plumbing specialist, to conduct an immediate and thorough investigation into the precise source of the water leakage affecting the bedroom ceiling. Following identification, rapid remediation of the source is paramount. Concurrently, a certified mold remediation specialist must be contracted without delay to safely and completely remove all detected mold growth and to restore any building materials compromised by moisture. These actions are nonnegotiable and require immediate commencement to mitigate further structural damage and eliminate health risks.

Urgent Action (Living Room):

A licensed structural engineer should be commissioned to perform a comprehensive evaluation of the wall cracks in the living room, accurately determining their origin, assessing their structural significance, and recommending appropriate stabilization or repair strategies. Simultaneously, a certified building envelope specialist or energy auditor should conduct an indepth analysis of the thermal anomalies detected, pinpointing the exact sources of heat leakage and temperature variation,

and proposing targeted solutions for improving thermal performance and air sealing. These investigations and initial remedial steps should be prioritized within a short timeframe.

General Recommendations:

Upon successful remediation of the root causes and any necessary structural repairs, a professional painting contractor should be engaged to meticulously prepare and restore all affected wall and ceiling surfaces in both the living room and bedroom. This process should include proper surface preparation and the application of highquality, appropriate paint systems to ensure durability and prevent recurrence of cosmetic issues.

6. ADDITIONAL NOTES

The convergence of visible physical deterioration and corroborating thermal anomalies across multiple areas of the property strongly suggests that the building may be experiencing systemic issues related to its overall envelope integrity and internal moisture management. To provide a comprehensive safeguard for the property and its occupants, a more expansive, holistic assessment of the entire structure is highly recommended. This broader evaluation, extending beyond the immediately identified problem areas, would be invaluable in preemptively identifying other vulnerable points, thereby ensuring the longterm structural stability, energy efficiency, and occupant comfort of the entire residence. Furthermore, all proposed remedial actions should be executed in strict adherence to current local building codes and industry best practices concerning moisture management, structural repairs, and indoor air quality standards. The efficacy of integrating thermal diagnostics with traditional visual inspections has proven instrumental in substantiating the observed defects and guiding targeted interventions.

7. MISSING OR UNCLEAR INFORMATION

While this diagnostic report provides a robust initial assessment, certain critical data points are currently unavailable, which would significantly enhance the precision and scope of future remediation planning. Specifically, the exact source and prolonged history of the water leakage in the bedroom remain undetermined; a more detailed and potentially intrusive water intrusion investigation is essential to pinpoint the precise entry point and mechanism. The full structural implications of the wall cracks in the living room require further quantitative engineering evaluation, including precise measurements of crack widths, depth, and ongoing monitoring for any potential movement. Furthermore, comprehensive information regarding the specific construction materials, the exact age of the building, and its detailed maintenance history, including any previous repair work, is currently not available. Obtaining this supplementary information, potentially through additional investigative methods such as moisture mapping, exploratory demolition, and detailed interviews with the property owner, would provide crucial context, enable a more definitive diagnosis, and facilitate the development of a highly targeted and effective longterm repair strategy.

EXTRACTED IMAGES

Inspection Report Images:

Page 1: inspection_page_1_img_1.jpeg

Thermal Report Images:

Page 1: thermal_page_1_img_1.jpeg

APPENDIX

■ Premium Report Generation Information

This sophisticated Detailed Diagnostic Report has been generated using cutting-edge AI technology combined with advanced building science principles. Our system represents the pinnacle of automated property analysis, delivering insights that traditionally required extensive manual expertise and time investment.

■ Advanced Technical Specifications

• **Analysis Engine:** Google Gemini 2.5 Flash - State-of-the-art AI model
• **PDF Processing:** PyMuPDF Advanced - High-fidelity document extraction
• **Report Generation:** Professional ReportLab Engine - Premium document formatting
• **Processing Date:** 2026-03-18 22:59:38
• **Report Version:** DDR Premium v2.0
• **Quality Assurance:** Multi-layer validation and verification

■ Our Commitment to Excellence

Every report generated through our system undergoes rigorous quality checks and validation processes. We combine the power of artificial intelligence with proven building science methodologies to deliver reports that professionals can trust and clients can understand.

■ Data Processing Capabilities

Our system processes multiple data streams simultaneously, including visual inspection data, thermal imaging information, and environmental factors. This comprehensive approach ensures that no critical information is overlooked and that all findings are properly contextualized within the broader property assessment framework.

■ Continuous Improvement

Our AI models are continuously trained on the latest building science research, industry standards, and real-world inspection data. This ensures that every report reflects the most current understanding of building performance, material science, and diagnostic methodologies.

Thank you for choosing our premium DDR analysis service. We are committed to providing the highest quality property assessments available in the industry today.